Large-Radius Local Flatness Hypothesis: Observable Radius, Curvature Limits, and Horizon Chaining

A conservative technical note / theoretical audit of the main geometric roadmap

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Abstract

This note consolidates the main, non-navigation roadmap of the Large-Radius Local Flatness Hypothesis. The core statement is deliberately conservative: local spatial or spacetime measurements may be insensitive to global curvature when the local scale L is much smaller than a curvature or horizon scale R, with leading geometric corrections of order $O((L/R)^2)$. The note separates local flatness, observational flatness, global flatness, topology, and total universe size. It also records a model-dependent flat-ΛCDM calculation of the observable radius, obtaining an illustrative particle-horizon scale of 46.13 Gly and observable diameter 92.27 Gly for Planck-like parameters. The Big Bang is explicitly rejected as a spatial center. The "horizon chaining" idea — placing ideal observers near our observable boundary and considering their own observable spheres — is retained as a useful conditional thought experiment, not a measurement of total universe size. No new observational fit, no claim of global topology, no gravity/dark-matter/dark-energy derivation, and no reopening of the earlier membrane branch are made.

Status and scope

This document is intended as a public technical note suitable for Zenodo deposition and written in an arXiv-like style. It consolidates V1024–V1037 and V1052–V1054 of the main cosmological/geometric branch. It deliberately excludes the separate space-navigation prototype branch, which may later be treated as a distinct software or educational toolkit.

Boundary. This is not a peer-reviewed cosmology paper and does not claim discovery of a new cosmological model. It is a structured audit of what can be stated safely from the project roadmap.

1 Motivation

The motivating intuition is simple: a sufficiently small local domain may appear flat even if the global geometry is curved, just as a small patch of a large sphere appears approximately planar. The project goal was not to replace general relativity or infer new physics, but to formalize the local-limit statement and determine which stronger claims must remain blocked.

The earlier 2D/3D membrane routes are historical context only. They did not produce a defensible physical-gravity gate and are not reopened here. The present branch is instead a conservative geometry-and-horizon audit.

2 Geometric local-flatness core

Let L be the scale of a local experiment, patch, or observational domain, and let R denote an appropriate curvature scale. In normal coordinates around a point or along a local observer

worldline, metric departures from the locally flat form enter at quadratic order in distance. Schematically,

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + O\left(\frac{x^2}{R^2}\right), \tag{1}$$

so a local dimensionless correction at scale L has the form

$$\epsilon_{\text{local}} \sim O\left(\frac{L^2}{R^2}\right). \tag{2}$$

This is the main defensible result of the local-flatness branch: it is a geometric approximation statement, not a physical explanation of gravity or cosmological acceleration.

3 FLRW curvature dictionary

For an FLRW spatial curvature parameter Ω_k, the curvature radius can be expressed as

$$R_k = \frac{R_H}{\sqrt{|\Omega_k|}}, \qquad R_H = \frac{c}{H_0}, \tag{3}$$

when $\Omega_k \neq 0$. Hence the local flatness scale for a selected tolerance ϵ can be represented conditionally as

$$D_{\text{flat}}(\epsilon) = R_k\sqrt{\epsilon}. \tag{4}$$

If $\Omega_k = 0$ exactly in the FLRW spatial-curvature sense, this curvature-threshold route formally diverges, but this does not decide global topology or total spatial size.

Planck 2018 gives a standard reference parameter set for base ΛCDM, including $H_0 = (67.4 \pm 0.5)\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ and $\Omega_m = 0.315 \pm 0.007$ [1]. Combining Planck with BAO gives curvature constraints compatible with spatial flatness; this supports the conservative use of large curvature radii but does not measure a total universe size.

4 Big Bang is not a spatial center

A key audit result is conceptual: the Big Bang cannot be used as the central point from which to measure the radius of the universe. In standard cosmology the Big Bang is not an explosion at one place in pre-existing empty space; it is a hot dense early state of the universe and an origin/limit of the cosmic expansion history. NASA education material emphasizes that the observable universe and the cosmic microwave background are understood as all-sky relics of the early universe rather than as light from a central explosion point [3, 4].

Therefore the following inference is rejected:

$$\text{Big Bang as spatial center} \quad \Rightarrow \quad \text{total universe radius.} \tag{5}$$

The valid calculation is instead an observer-centered horizon calculation using $H(z)$.

5 Observable radius calculation

For a flat ΛCDM illustrative model,

$$E(z) = \frac{H(z)}{H_0} = \sqrt{\Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_\Lambda}, \tag{6}$$

with $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$. The comoving particle horizon is

$$R_{\text{obs}} = c\int_0^\infty \frac{dz}{H(z)}. \tag{7}$$

Equivalently, in scale factor a,

$$R_{\rm obs} = R_H \int_0^1 \frac{da}{a^2 E(a)}. \tag{8}$$

This gives the present comoving radius of the region from which light signals could have reached us since the hot early universe. It is not a light-travel-time distance and it is not the total size of the universe. NASA gives the observable-universe diameter as about 92 billion light-years in public outreach material [2].

Table 1: Illustrative flat-ΛCDM distance scales reproduced in V1052/V1055C. These are model calculations from declared parameters, not new fits.

Quantity	Value	Unit
H_0	67.4	km s^{-1} Mpc^{-1}
Ω_m	0.315	dimensionless
Ω_r	9.2×10^{-5}	dimensionless
Ω_Λ	0.684908	dimensionless
Hubble radius c/H_0	14.507	Gly
Integrated age	13.791	Gyr
CMB comoving distance, $z = 1089.9$	45.219	Gly
Observable radius / particle horizon	46.133	Gly
Observable diameter	92.267	Gly
Illustrative event horizon	16.679	Gly

6 Observable versus total universe size

The observable radius is centered on the observer. It is not a boundary of the whole universe. Another observer far away has a different observable sphere, with partial overlap or no overlap depending on separation and cosmic history. Thus the project keeps three levels separate:

$$R_{\rm obs} \approx 46.13\,\text{Gly} \qquad \text{model-dependent observable radius,} \tag{9}$$

$$R_k = R_H/\sqrt{|\Omega_k|} \qquad \text{conditional curvature radius,} \tag{10}$$

$$R_{\rm total} \quad \text{unknown} \qquad \text{depends on global topology/extension.} \tag{11}$$

A flat universe need not be infinite unless additional assumptions such as simple connectedness are imposed. A compact flat topology, such as a 3-torus, is locally flat but globally finite. Searches for matched circles and other topology signatures can constrain some compact topologies, but absence of a detected topology does not provide a complete total-size measurement [6].

7 Horizon chaining

The user-proposed horizon-chaining intuition is retained as a useful thought experiment. Let our present comoving observable radius be

$$R_{\rm obs} \simeq 46.13\,\text{Gly}. \tag{12}$$

Along one idealized line through us, our observable interval is

$$[-R_{\rm obs}, +R_{\rm obs}], \tag{13}$$

with diameter $2R_{\rm obs} \simeq 92.27\,\text{Gly}$. If ideal comoving observers existed near both ends of our observable interval and if the spatial manifold continued without topological wrap or boundary,

each would have its own observable interval. The union along that one-dimensional line becomes schematically

$$[-2R_{\mathrm{obs}}, +2R_{\mathrm{obs}}], \tag{14}$$

with length

$$4R_{\mathrm{obs}} \simeq 184.53\,\mathrm{Gly}. \tag{15}$$

This does not prove that the universe is at least 184.53 Gly across in every global sense. It demonstrates instead that the edge of our observable universe is not automatically the edge of the universe. Iterating the argument suggests the possibility of arbitrarily large or infinite extension, but only under assumptions about continuation and topology that are not observed directly.

8 Claim matrix

Table 2: Main claim boundary.

Claim	Status	Public wording
Local flatness scales as $O((L/R)^2)$	allowed	Geometric local-limit statement.
$R_k = R_H/\sqrt{\lvert\Omega_k\rvert}$	allowed	Standard FLRW curvature dictionary when $\Omega_k \neq 0$.
Observable radius from $H(z)$	allowed as model calculation	Illustrative flat-ΛCDM calculation; no fit.
Observable diameter around 92 Gly	allowed	Observable universe only.
Big Bang as spatial center	rejected	No spatial center is used or inferred.
Total universe size measured	rejected	Unknown without global topology/extension assumptions.
Flat implies infinite	rejected	Flat compact topologies remain logically possible.
Horizon chaining equals total size	rejected	Conditional thought experiment only.
Gravity/DM/DE derivation	rejected	No physical-gravity gate opened.

9 Roadmap coverage

- V1024–V1026: preregistration, normal-coordinate derivation, FLRW curvature dictionary.
- V1027–V1032: observable matrix, scale hierarchy, public cleanup, geometry/observation audits.
- V1033: first-cycle recovery and branch separation.
- V1034–V1037: observable versus total universe size, horizon definitions, conditional size scenarios, topology/compactness preregistration.
- V1052: Big Bang not-center and observable-radius calculation audit.
- V1054: horizon chaining retained as a conditional thought experiment; navigation content excluded from this synthesis.

10 Conclusion

The safe outcome of the roadmap is a conservative geometric thesis: local flatness is expected when $L \ll R$, with corrections of order $O((L/R)^2)$; the observable universe has a model-dependent

present comoving radius of about 46 Gly and diameter about 92 Gly; the Big Bang is not a spatial center; and horizon chaining shows why the observable boundary is not the total boundary. The total universe size remains unknown. No new gravity, dark matter, dark energy, topology, or membrane claim is made.

Reproducibility note

The package includes the script `src/v1055b_observable_radius_calculation.py`, numeric output table `tables/v1055b_observable_radius_values.csv`, lineage index, claim matrix, status JSON, and SHA256 manifest.

References

[1] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astronomy & Astrophysics* 641, A6 (2020). Also arXiv:1807.06209. https://arxiv.org/abs/1807.06209

[2] NASA, "How Big is Space? We Asked a NASA Expert," NASA Science / Astrophysics outreach page. https://www.nasa.gov/science-research/astrophysics/how-big-is-space-we-asked-a-nasa-expert-episode-61/

[3] NASA GSFC, "Ask an Astrophysicist - Cosmology," Imagine the Universe! https://imagine.gsfc.nasa.gov/ask_astro/cosmology.html

[4] NASA, "WMAP Overview," NASA Science. https://science.nasa.gov/mission/wmap/wmap-overview/

[5] T. M. Davis and C. H. Lineweaver, "Expanding Confusion: common misconceptions of cosmological horizons and the superluminal expansion of the Universe," *Publications of the Astronomical Society of Australia* 21, 97–109 (2004). arXiv:astro-ph/0310808.

[6] J. S. Key, N. J. Cornish, D. N. Spergel, and G. D. Starkman, "Extending the WMAP Bound on the Size of the Universe," arXiv:astro-ph/0604616 (2006).